

Han Peng

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EDUCATION

2014 – 2018 Department of Physics, University of Oxford DPhil (PhD) Reading in Condensed Matter Physics

2010 – 2014 University of Science and Technology of China (USTC) Bachelor of Science in Applied Physics

RESEARCH

2018 – Present Postdoctoral project: Deep Medicine Advisor: Prof Steve Smith and Prof Andrei Vedaldi

Oxford Centre for Functional MRI and Brain, Nuffield Department of Clinical Neuroscience

AND Visual Geometry Group, Department of Engineering Sciences, University of Oxford

- **Deep learning in UK Biobank MRI data**

Looking for disease and aging patterns in ~20000 subjects with 3D MRI data sets

2014 – 2018 PhD Project: Study of Electronic Properties of Quantum Materials Advisor: Prof Yulin Chen

Clarendon Laboratory, Department of Physics, University of Oxford

- **Machine learning in ARPES data analysis**

Used convolutional neural network to identify the dispersive features in the spectrum images. The project is developed using MATLAB, Python and TensorFlow

- **Twisted bilayer graphene**

Used Angular Resolved Photo-Emission Spectroscopy (ARPES) with Micrometre resolution to study electronic properties of twisted bilayer graphene

- **Other contributions**

Led the development of data analysis system for spectrum image data processing.

There are 30 users, 4 regular contributors and 10 interns throughout the years. The work includes algorithm optimisation, data visualisation and graphic user interface building

2017 Research Intern: Neuron Image Registration Advisor: Prof Kenneth Harris

Cortical Processing Laboratory, University College London

- Designed algorithm to register ex-vivo 2D-slices with in-vivo 3D-volume data of neuronal images with different modality and detection methods

- Developed semi-automatic MATLAB data analysis pipeline for the registration algorithm. The pipeline includes data I/O, cell detection, registration, result visualisation and manual modification interface. Project GitHub page: <https://git.io/vAjk4>

2013 Research Intern: Study of Jamming Transition in Soft Materials Advisor: Prof Ning Xu

Computational and Theoretical Soft Condensed Matter Physics Laboratory, USTC

- Computationally studied the power law phenomenon in the Jamming transition of a particle system
- Analysed the performance of energy minimization algorithm in different energy potential field

TEACHING

2015 – 2018 Demonstrator (Senior Demonstrator since 2017) in **Computational Physics**, University of Oxford

2017 – 2018 Tutor in **Mechanics**, Jesus College, University of Oxford

2016 – 2017 Tutor in **Condensed Matter Physics**, Hertford College, University of Oxford

AWARDS AND FUNDING

2014 – 2018 Oxford University – China Scholarship Council Scholarship, full scholarship for PhD study

2015 Arthur H. Cooke Memorial Prize awarded by Department of Physics, University of Oxford

TALKS AND PRESENTATIONS

2018.03 Van hove singularity evolution with twist angle in twisted bilayer graphene

APS March meeting 2018, Los Angeles, CA, US

2017.11 Register 2D neuron image with 3D data

Neuromics project annual retreat 2017, Washington D.C, US

PUBLICATIONS

Total citation: 1472 | h-index: 7 (Nov 2018) | Google Scholar page: <https://goo.gl/vINvHL>

Data analysis

1. **H. Peng**, X. Gao, Y. He, Y. W. Li, Y. C. Ji, C. H. Liu, S. A. Ekahana, D. Pei, Z. K. Liu, Z. X. Shen, Y. L. Chen "Super Resolution Convolutional Neural Network for Feature Extraction in Spectroscopic Data" *Manuscript ready*
2. C. Chen, M. X. Wang, J. X. Wu, Y. Sun, H. F. Yang, Z. Tian, T. Tu, **H. Peng**, G. Li, H. X. Fu, X. Xu, J. Jiang, N. B. M. Schroeter, Y. W. Li, D. Pei, S. Liu, S. Ekahana, H. T. Yuan, J. M. Xue, Z. K. Liu, B. H. Yan, H. L. Peng and Y. L. Chen "Electronic Structures of a High-Mobility Layered Oxychalcogenide Semiconductor, Bi₂O₂Se" *Science Advances*, **4** (9), eaat8355 (2018)
3. S. L. Zhang, W. W. Wang, D. M. Burn, **H. Peng**, H. Berger, A. Bauer, C. Pfleiderer, G. van der Laan, T. Hesjedal "Manipulation of skyrmion motion by magnetic field gradients" *Nature Communications*, **9** (1), 2115 (2018)

Graphene

4. H. F. Yang, C. Chen, H. Wang, Z. K. Liu, T. Zhang, **H. Peng**, N.B.M. Schröter, S. A. Ekahana, J. Jiang, L. X. Yang, V. Kandyba, A. Barinov, C. Y. Chen, J. Avila, M. C. Asensio, H. L. Peng, Z. F. Liu, and Y. L. Chen "Single Crystalline Electronic Structure and Growth Mechanism of Aligned Square Graphene Sheets" *APL Materials*, **6** (3), 036107 (2018)
5. **H. Peng**[†], N. B. M. Schröter[†] (equal contribution), J. B. Yin, H. Wang, T.-F. Chung, H. F. Yang, S. Ekahana, Z. K. Liu, J. Jiang, L. X. Yang, T. Zhang, C. Chen, H. Ni, H. Barinov, Y. P. Chen, Z. F. Liu, H. L. Peng, Y. L. Chen "Substrate Doping Effect and Unusually Large Angle van Hove Singularity Evolution in Twisted Bi- and Multilayer Graphene" *Advanced Materials* (2017)
6. J. B. Yin[†], H. Wang[†], **H. Peng**[†] (equal contribution), Z. J. Tan, L. Liao, L. Lin, X. Sun, A. L. Koh, Y. L. Chen, H. L. Peng, and Z. F. Liu "Selectively enhanced photocurrent generation in twisted bilayer graphene with van Hove singularity" *Nature Communications*, **7**, 10699 (2016)
7. L. Liao, H. Wang, **H. Peng**, J. B. Yin, A. L. Koh, Y. L. Chen, Q. Xie, H. L. Peng, and Z. F. Liu "van Hove Singularity Enhanced Photochemical Reactivity of Twisted Bilayer Graphene" *Nano Letters*, **15**, 5585 (2015)

Other ARPES related topics

8. J. Jiang, N. B. M. Schröter, S.-C. Wu, N. Kumar, C. Shekhar, **H. Peng**, X. Xu, C. Chen, H. F. Yang, C.-C. Hwang, S.-K. Mo, C. Felser, B. H. Yan, Z. K. Liu, L. X. Yang, and Y. L. Chen "Observation of topological surface states and strong electron/hole imbalance in extreme magnetoresistance compound LaBi" *Physical Review Materials*, **2**, 024201 (2018)
9. A. J. Liang, J. Jiang, M. X. Wang, Y. Sun, N. Kumar, C. Shekhar, C. Chen, **H. Peng**, C. W. Wang, X. Xu, H. F. Yang, S. T. Cui, G. H. Hong, Y. Y. Xia, S. K. Mo, Q. Gao, X. J. Zhou, L. X. Yang, C. Felser, B. H. Yan, Z. K. Liu, Y. L. Chen 2017. Observation of the topological surface state in the nonsymmorphic topological insulator KHgSb. *Physical Review B*, **96**(16), p.165143 (2017)
10. J. Jiang, Z. K. Liu, Y. Sun, H. F. Yang, R. Rajamathi, Y. P. Qi, L. X. Yang, C. Chen, **H. Peng**, C.-C. Hwang, S. Z. Sun, S.-K. Mo, I. Vobornik, J. Fujii, S. S. P. Parkin, C. Felser, B. H. Yan, Y. L. Chen "Signature of type-II Weyl semimetal phase in MoTe₂" *Nature Communications*, **8**, 13973 (2017)
11. Z. K. Liu, L. X. Yang, Y. Sun, T. Zhang, **H. Peng**, H. F. Yang, C. Chen, Y. Zhang, Y. F. Guo, D. Prabhakaran, M. Schmidt, Z. Hussain, S.-K. Mo, C. Felser, B. Yan and Y. L. Chen "Evolution of the Fermi surface of Weyl semimetals in the transition metal pnictide family" *Nature Materials*, **15**, 27 (2016)
12. L. X. Yang, Z. K. Liu, Y. Sun, **H. Peng**, H. F. Yang, T. Zhang, B. Zhou, Y. Zhang, Y. F. Guo, M. Rahn, D. Prabhakaran, Z. Hussain, S.-K. Mo, C. Felser, B. Yan and Y. L. Chen "Weyl Semimetal Phase in non-Centrosymmetric Compound TaAs" *Nature Physics*, **11**, 728 (2015)
13. Z. K. Liu, J. Jiang, B. Zhou, Z. J. Wang, Y. Zhang, H. M. Weng, D. Prabhakaran, S. -K. Mo, **H. Peng**, P. Dudin, T. Kim, M. Hoesch, Z. Fang, X. Dai, Z. X. Shen, D. L. Feng, Z. Hussain, Y. L. Chen. "A Stable Three-dimensional Topological Dirac Semimetal Cd₃As₂" *Nature Materials*, **13**, 677 (2014)

REFERENCE

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Prof Thorsten Hesjedal Physics Department, University of Oxford Thorsten.Hesjedal@physics.ox.ac.uk